

Searching - Algorithm

All - Questions

Question # 01

```
int BS(int arr[], int size, int t)
{
    for(int i=0; i<size; i++)
    {
        if(arr[i]==t)
        {
            return i;
        }
    }
    return -1;
}
```

(DRY - RUN)

Suppose:

arr is {1, 2, 7, 9}
target is 7
size = 4

1) for(int i=0; 0<4; i++) ✓
{
if(arr[i]==t) X 1 ≠ 7

2) for(→) 0<4 ✓
{
if(→) X 2 ≠ 7

3) for(→) ✓ 2<4
if(→) ✓ 7=7
return i; i=2

Question # 02

// Just loop $\rightarrow$ Explained in DRY RUN
(DRY_RUN)

Suppose: arr is = {1, 2, 3, 4, 5}
target = 4
size = 5

int left = 0;

int Right = size - 1;

15 - 4 = 4 R = 4

1) while (L <= R) ✓
{

int mid = L + (R - L) / 2;

0 + 4 / 2 = 0 + 2 = 2
mid = 2

if (arr[mid] == target) ✗

if (arr[mid] < target) ✓
{

L = mid + 1;

L = 3

2) again while (L <= R)
{

int m = L + (R - L) / 2;

3 + 1 / 2 =
mid = 3

if (arr[mid] == target) ✓ 4 = 4
{
return mid;
}

Question # 03

// Just loop $\rightarrow$ Explained in dry-run

(DRY-RUN)

$\rightarrow 0 \rightarrow R = \text{size} - 1 = 6 - 1 = 5$

① int bsr(int arr[], int L, int R, int T)

{
if (L > R) $\times$ $L=0, R=5$

int mid = $L + (R - L) / 2$; $\frac{0+5}{2}$
 $\boxed{\text{mid}=2}$

Suppose arr is
{1, 2, 3, 4, 5, 6}
target is 4
size = 6

if (arr[mid] == target) $\times$

if (arr[mid] < target) $\checkmark$

return bsr(arr, mid+1, R, T);

now
return to
main or
caller

② recursing

bsr(int* arr, int L, int R, int T)

$\rightarrow 3 \rightarrow 5 \rightarrow 7$

if (L > R) $\times$

int mid = $L + (R - L) / 2$; $\frac{3+7}{2} =$
mid = 4

if (arr[mid] == T) $\checkmark$

{ return mid; } // return to caller ①